

Learning how to prove Workshop Week 10

Lagrange's Theorem



Cosets. Lagrange's Theorem.

1. Find the right cosets in G of the subgroup H of G .
 - (a) $G = C_{20} = \langle a \rangle$, $\text{o}(a) = 20$ and $H = \langle a^4 \rangle$.
 - (b) $G = \mathbb{Z}_{12}$, $H = \langle \bar{4} \rangle = \{\bar{0}, \bar{4}, \bar{8}\}$.
 - (c) Let G be a group and $a \in G$. Suppose that a has order 15. Find all the (distinct) right cosets of $\langle a^5 \rangle$ in $\langle a \rangle$.

2. Let K and H subgroups of a group G and $g \in G$. Show that $g(H \cap K) = gH \cap gK$.